

How does gender affect hitchhiking: Does gender influence vehicle type?

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Introduction

Hitchhiking is a form of transportation in which individuals seek free rides from passing drivers, typically by standing alongside roads. It is an intrinsically unpredictable activity, as the likelihood of being picked up depends on various factors, including chance and driver preferences [1].

This programming project investigates the scientific question: **Is hitchhiker gender associated with the type of vehicle providing transportation?** To investigate this question, a publicly available dataset stored on the website *huggingface.co*, but created by *hitchwiki.org*, was used. The dataset contains self-reported hitchhiking experiences, allowing individual hitchhikers to document their journeys and the characteristics of the drivers who picked them up.

Data sample

The data set used [2] in this study contained 84,700 entries from 2005 to 2026. Each entry contained 15 columns. The information per entry ranged from the information about the hitchhiker, the driver and the overall information about the hitchhikers experience.
To answer the question of this study, the data was cleaned so that only the following columns remained:

- Gender of hitchhiker
- Gender of driver
- Pick up vehicle (car, truck, van, bus, camper and others)

Furthermore, entries where these were not included were removed, but also entries with more than one hitchhiker, or more than one driver were removed. Adding the gender is a new feature of hitchwiki.org, hence, only entries from 2026 were used, resulting in 277 entries.

Results and discussion

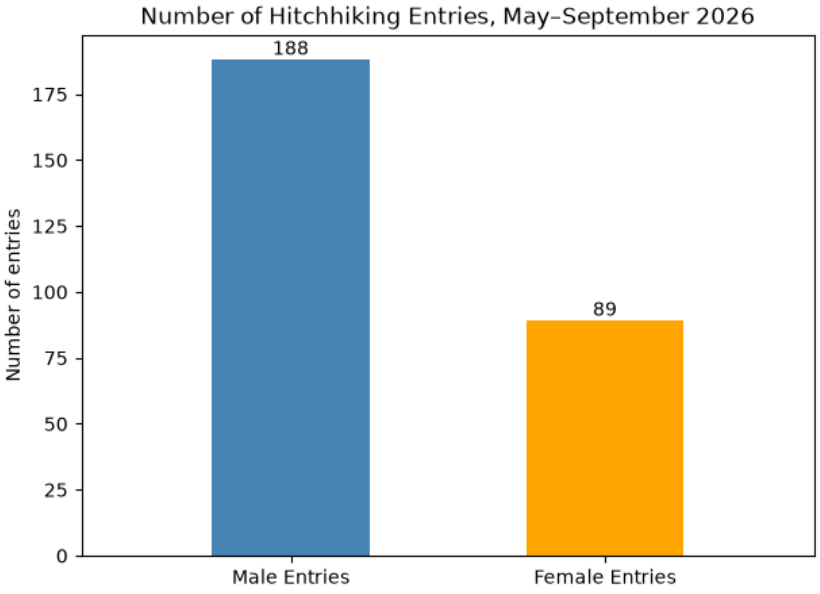


Figure 1: *Number of entries among male and female hitchhikers between May and September 2026.*

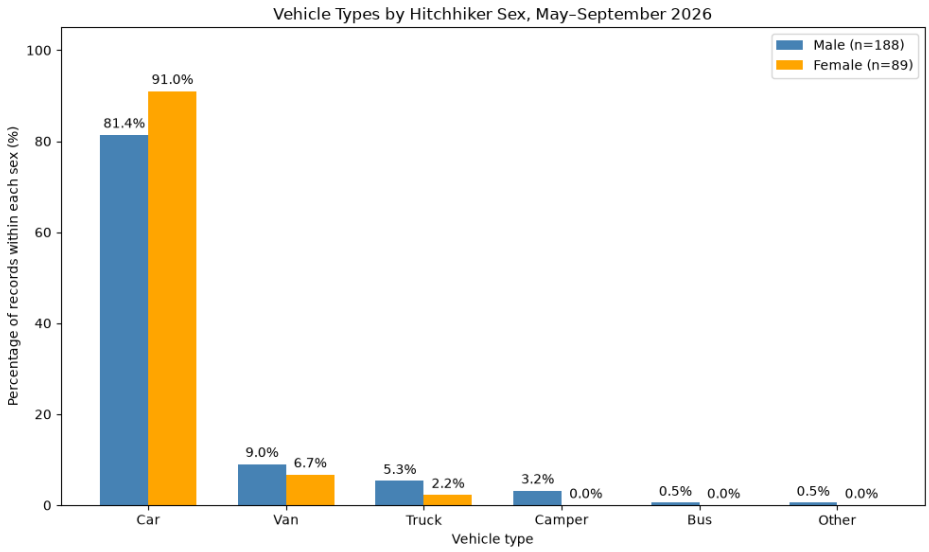


Figure 2: *Percentage distribution of vehicle types among male and female hitchhikers. Bars represent the proportion of each vehicle type within each gender. Cars were the most frequently recorded vehicle type for both genders.*

To investigate whether hitchhiker gender was associated with the type of vehicle providing transport, a contingency table was constructed using 277 hitchhiking entries recorded between May and September 2026, as shown below.

	Car	Van	Truck	Camper	Bus	Other	Total
Male	153	17	10	6	1	1	188
Female	81	6	2	0	0	0	89
Total	234	23	12	6	1	1	277

Of these, 188 entries involved male hitchhikers and 89 involved female hitchhikers.

Cars represented the most frequently recorded vehicle type for both genders, accounting for 81.4% of male entries and 91.0% of female entries. Vans were the second most common vehicle type, followed by trucks. Campers, buses, and other vehicle types were recorded exclusively among male hitchhikers, although their overall frequencies were low.

A Pearson chi-squared test of independence was performed to examine the association between hitchhiker gender and vehicle type. Due to the low expected frequencies in several vehicle categories, a Monte Carlo simulation with 10,000 replicates was used to estimate the p-value instead of relying on the theoretical chi-squared distribution. The test yielded $\chi^2 = 6.15$, p-value $= 0.2752$, indicating no statistically significant association according to the conventional asymptotic test.

The observed distribution suggests that cars were the predominant vehicle type for both male and female hitchhikers. Although the proportions of certain vehicle types differed between genders, the low frequencies of less common vehicle categories limit the strength of conclusions that can be drawn from these differences.

References

[1] Wenke, T. (2025). *Hitchhiking Rides Dataset: Two decades of crowd-sourced records on stochastic traveling*. arXiv preprint arXiv:2506.21946. Available at: https://arxiv.org/abs/2506.21946.

[2] Hitchwiki. *Hitchhiking Rides Dataset*. Hugging Face Datasets. Available at: https://huggingface.co/datasets/Hitchwiki/hitchhiking-rides-dataset. Accessed: 12 October 2026.